Monthly Child Tax Credit Payments Reduced Food Hardship, Most for Families With the Least

Evidence from the Census Household Pulse Survey, 2021–2022

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Key findings

- While the monthly payments were arriving (July–December 2021), the share of families with children who **sometimes or often did not have enough to eat fell by about 2.4 to 3.1 percentage points** compared with families without children. That is roughly a **one-fifth drop**.
- The drop was largest for families earning under $35,000 (about 7 points), Hispanic families, families with three or more children, and families without a college graduate. Families earning $75,000–$150,000 saw almost no change.
- The improvement faded in the months after the payments stopped and was gone by spring 2022.

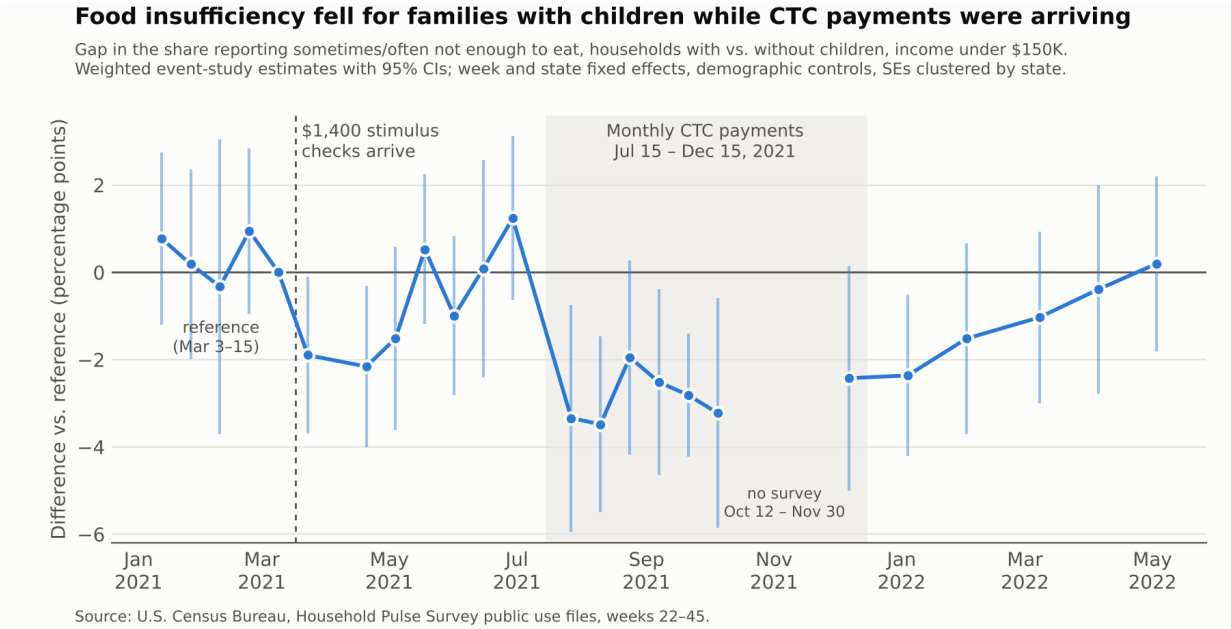
Background

For 2021 only, the American Rescue Plan Act raised the Child Tax Credit from $2,000 per child to $3,600 for each child under 6 and $3,000 for each child aged 6 to 17. It also made the credit fully refundable, so families with little or no earnings could receive the full amount; under prior law, the credit grew with earnings. For the first time, the IRS paid half of the credit in advance, as monthly deposits of up to $300 per young child and $250 per older child, from July 15 to December 15, 2021. Families received the rest when they filed their 2021 taxes. The expansion expired after 2021, and no monthly payments followed.

This brief asks two questions: did the monthly payments reduce food hardship, and which families gained the most?

Finding 1: Food hardship fell as soon as payments began, then returned after they ended

The Census Bureau's Household Pulse Survey asked adults every few weeks whether their household had enough to eat in the past seven days. I compared families with children, who received the payments, with families without children, who did not, before, during, and after the payment months. The figure shows how the gap between the two groups moved over time.

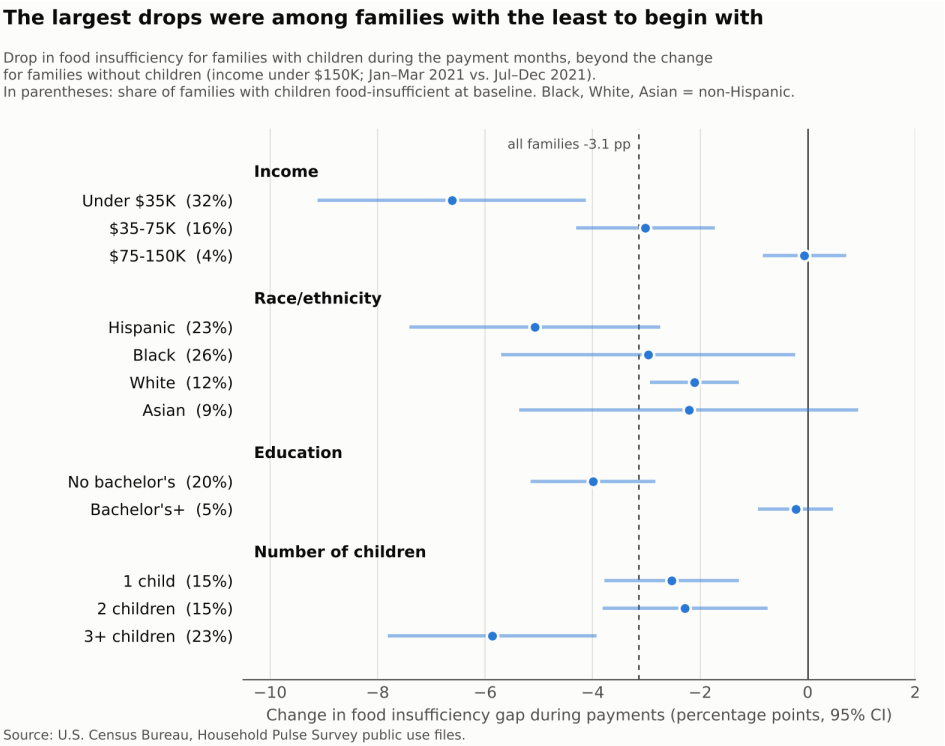


Each point compares households with and without children (income under $150,000) to the same comparison in early March 2021. Lower values mean families with children were doing relatively better. Vertical lines show 95 percent confidence intervals.

Before mid-March 2021, the gap was steady. It dipped when the $1,400-per-person stimulus checks arrived (those checks also paid families for each child) and returned to its earlier level by early summer. In the first survey after the first CTC payment in July, the gap dropped sharply, and it stayed lower through December. Across the payment months, food insufficiency among families with children fell by 2.4 to 3.1 percentage points relative to families without children, depending on which earlier period is used as the baseline. Before the payments, about 14 percent of these families were food-insufficient. After the payments ended, the gap closed gradually and was back at its pre-payment level by May 2022.

Finding 2: Families with the least gained the most

The biggest improvements went to families who were most likely to go without enough food to begin with. Among families earning under $35,000, about one in three was food-insufficient before the payments; the payments lowered that by roughly 7 percentage points. The drop was also large for Hispanic families (5 points), families with three or more children (6 points, consistent with larger payments for larger families), and families without a college graduate (4 points). Families earning $75,000–$150,000, and families with a college graduate, saw almost no change, likely because few of them were struggling to afford food in the first place.



Each point is a separate comparison within one group. Points further left mean a larger drop in food hardship. Lines show 95 percent confidence intervals.

A machine-learning method (a causal forest) that considers all of these characteristics together reached the same conclusion. I tested it on a half of the data it had not seen. The quarter of families it predicted would benefit most showed a 5-point drop, while the other three quarters showed little change. Most families in that top quarter had incomes under $35,000.

What this means for policy

- **Regular cash reached families quickly.** The drop appeared in the first survey after the first deposit, within about three weeks. For families on the edge, the timing of support matters as well as its size.
- **Full refundability shaped who benefited.** The largest gains went to the lowest-income families, who would have received a partial credit or none under the prior earnings-based rules. Decisions about how the credit phases in with earnings largely determine its effect on food hardship.
- **Temporary support brought temporary relief.** Food hardship returned within months of the last payment. Policymakers weighing short-term versus ongoing support should expect the effects to last about as long as the

payments do.

- **This brief does not weigh everything that matters.** It does not assess cost, effects on work, or other outcomes such as housing or health, which other studies have examined and which belong in any full assessment.

How confident can we be?

The results line up with earlier studies using the same survey (Shafer et al., 2022; Parolin et al., 2021). When I repeated their comparisons, I found similar effects: a 4.5-point drop between the surveys just before and just after the first payment, versus their 3.7, and a 7.0-point drop for families under $35,000, versus their 7.5. The main estimate holds with all income levels included and with the Census Bureau's recommended method for calculating survey uncertainty. A similar drop shows up in another measure, difficulty paying usual household expenses.

There are limits. The survey asks different people each round, response rates are low, and answers are self-reported. Families with and without children may have been affected differently by other changes in 2021, such as expanded SNAP benefits, free school meals, and the end of extra unemployment benefits in September. The extended data from early 2021 help rule out the stimulus checks as an explanation, but not every other policy change. The income question also changed from 2019 to 2020 income just as the payments began; results that include all incomes are similar.

Data and methods

U.S. Census Bureau Household Pulse Survey public use files, weeks 22–45 (January 2021 to May 2022), 1.69 million adult responses. Main sample: households with incomes under $150,000 (761,268 responses). Weighted difference-in-differences and event-study regressions with survey-week and state fixed effects and demographic controls; standard errors are clustered by state and checked with replicate weights. Heterogeneity: subgroup regressions and causal forests validated on held-out data. Code and full results: github.com/ankitajaiwar/ctc-food-insufficiency.

References: Parolin, Z., Ananat, E., Collyer, S., Curran, M., & Wimer, C. (2021). The initial effects of the expanded Child Tax Credit on material hardship. NBER Working Paper 29285. Shafer, P. R., Gutiérrez, K. M., Ettinger de Cuba, S., Bovell-Ammon, A., & Raifman, J. (2022). Association of the implementation of Child Tax Credit advance payments with food insufficiency in US households. JAMA Network Open, 5(1), e2143296. IRS, 2021 Child Tax Credit frequently asked questions.